First, we learn that the familiar space and time that suffuse our thoughts and support our equations emerge from a kind of averaging process. Think of the pixelated image you see when your face is a few inches from a television screen. This image is very different from what you see at a more comfortable distance, because once you can no longer resolve individual pixels, your eyes combine them into an average that looks smooth. But notice that it's only through the averaging process that the pixels produce a familiar, continuous image. In a similar vein, the microscopic structure of spacetime is riddled with random undulations, but we aren't directly aware of them because we lack the ability to resolve spacetime on such minute scales. Instead, our eyes, and even our most powerful equipment, combine the undulations into an average, much like what happens with television pixels. Because the undulations are random, there are typically as many "up" undulations in a small region as there are "down," so when averaged they tend to cancel out, yielding a placid spacetime. But, as in the television analogy, *it's only because of the averaging process that a smooth and tranquil form for spacetime emerges.*

Quantum averaging provides a down-to-earth interpretation of the assertion that familiar spacetime may be illusory. Averages are useful for many purposes but, by design, they do not provide a sharp picture of underlying details. Although the average family in the U.S. has 2.2 children, you'd be in a bind were I to ask to visit such a family. And although the national average price for a gallon of milk is $2.783, you're unlikely to find a store selling it for exactly this price. So, too, familiar spacetime, itself the result of an averaging process, may not describe the details of something we'd want to call fundamental. Space and time may only be approximate, collective conceptions, extremely useful in analyzing the universe on all but ultramicroscopic scales, yet as illusory as a family with 2.2 children.

A second and related insight is that the increasingly intense quantum jitters that arise on decreasing scales suggest that the notion of being able to divide distances or durations into ever smaller units likely comes to an end at around the Planck length (10^{-33} centimeters) and Planck time (10^{-43} seconds). We encountered this idea in Chapter 12, where we emphasized that, although the notion is thoroughly at odds with our usual